Lecture 5: Inverse, Adjoint, and Projection Operators

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🔑 Learning Objectives

By the end of this lecture, you should be able to understand:

- Inverse operators
- Adjoint operators
- Projection operators (projectors)

Inverse Oprators

For one-to-one (injective) operators $\mathcal{H}$, a left inverse $\mathcal{H}_L^{-1}$ exists such that:

$$\mathcal{H}_L^{-1}\mathcal{H} = \mathcal{I}_{\mathbb{U}},$$

where $\mathcal{I}_{\mathbb{U}}$ is the identity operator in $\mathbb{U}$, mapping any vector to itself.

For onto (surjective) operators $\mathcal{H}$, a right inverse $\mathcal{H}_R^{-1}$ exists such that:

$$\mathcal{H}\mathcal{H}_R^{-1} = \mathcal{I}_{\mathbb{V}},$$

where $\mathcal{I}_{\mathbb{V}}$ is the identity operator in $\mathbb{V}$ mapping any vector to itself.

Adjoint Oprators

Let $\mathbf{f} \in \mathbb{U}$ and $\mathbf{g} \in \mathbb{V}$, where $\mathbb{U}$ and $\mathbb{V}$ are Hilbert spaces. The adjoint operator $\mathcal{H}^\dagger$ satisfies the relation:

$$\langle \mathbf{g}_2, \mathcal{H}\mathbf{f}_1 \rangle_{\mathbb{V}} = \langle \mathcal{H}^\dagger \mathbf{g}_2, \mathbf{f}_1 \rangle_{\mathbb{U}}.$$

Properties of Adjoint Operators

For a scalar $c \in \mathbb{C}$ and two different operators $\mathcal{H}_1$ and $\mathcal{H}_2$ mapping $\mathbb{U}$ to $\mathbb{V}$, the following properties hold:

- $(c\mathcal{H})^\dagger = c^* \mathcal{H}^\dagger$
- $(\mathcal{H}_1 + \mathcal{H}_2)^\dagger = \mathcal{H}_1^\dagger + \mathcal{H}_2^\dagger$
- $(\mathcal{H}_1 \mathcal{H}_2)^\dagger = \mathcal{H}_2^\dagger \mathcal{H}_1^\dagger$
- $(\mathcal{H}^\dagger)^\dagger = \mathcal{H}$

Special Cases

- Hermitian (self-adjoint): $\mathcal{H}^\dagger = \mathcal{H}$, where $\mathbb{U} = \mathbb{V}$
- Unitary: $\mathcal{H}^\dagger = \mathcal{H}^{-1}$, where $\mathbb{U} = \mathbb{V}$

$$\begin{aligned}
&\langle \mathcal{H}^\dagger \mathbf{g}_2, \mathbf{f}_1 \rangle_{\mathbb{U}} && \text{Inner product:} \langle \mathbf{a}, \mathbf{b} \rangle = \mathbf{a}^\dagger \mathbf{b} && (1)\\
&= (\mathcal{H}^\dagger \mathbf{g}_2)^\dagger \mathbf{f}_1 && (\mathcal{H}_1 \mathcal{H}_2)^\dagger = \mathcal{H}_2^\dagger \mathcal{H}_1^\dagger && (2)\\
&= (\mathbf{g}_2)^\dagger (\mathcal{H}^\dagger)^\dagger \mathbf{f}_1 && (\mathcal{H}^\dagger)^\dagger = \mathcal{H} && (3)\\
&= \mathbf{g}_2^\dagger \mathcal{H} \mathbf{f}_1 && && (4)\\
&= \langle \mathbf{g}_2, \mathcal{H}\mathbf{f}_1 \rangle_{\mathbb{V}} && && (5)
\end{aligned}$$

Example computation

Consider the Hilbert-space operator $\mathcal{H} : \mathbb{R}^2 \to \mathbb{R}^3$, represented by the matrix operator $\mathbf{H}$:

$$\mathbf{H} = \begin{bmatrix} 3 & 0 \\ 2 & 1 \\ -1 & -2 \end{bmatrix}, \quad \mathbf{f}_1 = \begin{bmatrix} 2 \\ -1 \end{bmatrix}$$

$$\mathbf{g}_1 = \mathbf{H}\mathbf{f}_1 = \begin{bmatrix} 3 & 0 \\ 2 & 1 \\ -1 & -2 \end{bmatrix} \begin{bmatrix} 2 \\ -1 \end{bmatrix} = \begin{bmatrix} 6 \\ 3 \\ 0 \end{bmatrix}$$

$$\mathbf{H}^\dagger = \begin{bmatrix} 3 & 2 & -1 \\ 0 & 1 & -2 \end{bmatrix}$$

$$\mathbf{H}^\dagger \mathbf{g}_1 = \mathbf{H}^\dagger \mathbf{H} \mathbf{f}_1 = \begin{bmatrix} 3 & 2 & -1 \\ 0 & 1 & -2 \end{bmatrix} \begin{bmatrix} 6 \\ 3 \\ 0 \end{bmatrix} = \begin{bmatrix} 24 \\ 3 \end{bmatrix} \neq \mathbf{f}_1$$

$$\mathbf{H}^\dagger \mathbf{H} \neq \mathbf{I}$$

$$\mathbf{H}^\dagger \neq \mathbf{H}^{-1} \quad \textbf{Not unitary}$$

In [12]:
```python
import numpy as np

# Define the matrix operator H
H = np.array([[3, 0],
              [2, 1],
              [-1, -2]])
print("H:\n", H)

# Define the vector f1
f1 = np.array([[2],
               [-1]])
print("f:\n", f1)

# Compute g1 = H f1
# g1 = np.matmul(H, f1)
g1 = H @ f1
print("g1 = H f1:\n", g1)
```

```
H:
 [[ 3  0]
 [ 2  1]
 [-1 -2]]
f:
 [[ 2]
 [-1]]
g1 = H f1:
 [[6]
 [3]
 [0]]
```

In [13]:
```python
# Compute the adjoint of H
H_adj = H.conj().T
print("H:\n", H)
print("H^† (Adjoint of H):\n", H_adj)

# Compute H_adj H
# H_adj_H = np.matmul(H_adj, H)
H_adj_H = H_adj @ H
print("H^† H:\n", H_adj_H)
```

```
H:
 [[ 3  0]
 [ 2  1]
 [-1 -2]]
H^† (Adjoint of H):
 [[ 3  2 -1]
 [ 0  1 -2]]
H^† H:
 [[14  4]
 [ 4  5]]
```

Example Use

Adjoint operators play a crucial role in optimization-based image reconstruction techniques, particularly in iterative methods.

Adjoint of a Matrix Operator (D-D mapping)

Let $\mathcal{H}$ be a matrix operator defined as:

$$\mathcal{H} : \mathbb{E}^N \to \mathbb{E}^M, \mathbf{H} \in \mathbb{E}^{M \times N}.$$

The adjoint of $\mathcal{H}$ is given by:

$$\mathcal{H}^\dagger : \mathbb{E}^M \to \mathbb{E}^N, \mathbf{H}^\dagger \in \mathbb{E}^{N \times M}.$$

The adjoint satisfies the relationship:

$$\langle \mathbf{g}_2, \mathcal{H}\mathbf{f}_1 \rangle_{\mathbb{V}} = \langle \mathcal{H}^\dagger \mathbf{g}_2, \mathbf{f}_1 \rangle_{\mathbb{U}}.$$

🧠 *Recall the inner product in a discrete form* (in Lecture 2 and Lecture 4):
$\langle \mathbf{f}_1, \mathbf{f}_2 \rangle = \sum_{n=1}^{N} f_{1n}^* f_{2n} = \mathbf{f}_1^\dagger \mathbf{f}_2$

Derivation of the Adjoint

$$\langle \mathbf{g}_2, \mathcal{H}\mathbf{f}_1 \rangle_{\mathbb{V}} = \langle \mathcal{H}^\dagger \mathbf{g}_2, \mathbf{f}_1 \rangle_{\mathbb{U}}$$

🧠 *Recall the D-D mapping, matrix operators* (in Lecture 4): $g_m = \sum_{n=1}^{N} H_{mn} f_n$

$$\langle \mathbf{g}_2, \mathcal{H}\mathbf{f}_1 \rangle_{\mathbb{V}} = \sum_{m=1}^{M} g_{2m}^* \sum_{n=1}^{N} H_{mn}\, f_{1n} = \sum_{n=1}^{N} \sum_{m=1}^{M} H_{mn}\, g_{2m}^*\, f_{1n}$$

$$\langle \mathcal{H}^\dagger \mathbf{g}_2, \mathbf{f}_1 \rangle_{\mathbb{U}} = \sum_{n=1}^{N} \Big[\sum_{m=1}^{M} [\mathbf{H}^\dagger]_{nm}\, g_{2m} \Big]^* f_{1n} = \sum_{n=1}^{N} \sum_{m=1}^{M} [\mathbf{H}^\dagger]_{nm}^*\, g_{2m}^*\, f_{1n}$$

$$[\mathbf{H}^\dagger]^*_{nm} = H_{mn}$$
$$[\mathbf{H}^\dagger]_{nm} = H^*_{mn} \quad \textbf{Conjugate transpose}$$

Adjoint of an Integral Operator (C-C mapping)

For an operator $\mathcal{H}$ mapping between Hilbert spaces of square-integrable functions:

$$\mathcal{H} : \mathbb{L}_2(\alpha, \beta) \to \mathbb{L}_2(\alpha, \beta).$$

The adjoint of $\mathcal{H}$ is defined as:

$$\mathcal{H}^\dagger : \mathbb{L}_2(\alpha, \beta) \to \mathbb{L}_2(\alpha, \beta)$$

The adjoint satisfies the inner product relationship:

$$\langle \mathbf{g}_2, \mathcal{H}\mathbf{f}_1 \rangle = \langle \mathcal{H}^\dagger \mathbf{g}_2, \mathbf{f}_1 \rangle$$

🧠 *Recall the inner product in a continuous form* (in Lecture 2):
$\langle \mathbf{f}_1, \mathbf{f}_2 \rangle = \int_\alpha^\beta dx f_1^*(x) f_2(x)$

Derivation of the Adjoint

$$\langle \mathbf{g}_2, \mathcal{H}\mathbf{f}_1 \rangle = \langle \mathcal{H}^\dagger \mathbf{g}_2, \mathbf{f}_1 \rangle$$

🧠 *Recall the C-C mapping, integral transforms* (in Lecture 4):
$g(x') = \int_\alpha^\beta dx\, h(x', x)\, f(x)$

$$\langle \mathbf{g}_2, \mathcal{H}\mathbf{f}_1 \rangle = \int_\alpha^\beta dx' g_2^*(x') \int_\alpha^\beta dx\, h(x', x) f_1(x) = \int_\alpha^\beta dx \int_\alpha^\beta dx' h(x', x) g_2^*(x') f_1(x)$$
$$\langle \mathcal{H}^\dagger \mathbf{g}_2, \mathbf{f}_1 \rangle = \int_\alpha^\beta dx \Big[\int_\alpha^\beta dx' h^\dagger(x, x') g_2(x') \Big]^* f_1(x) = \int_\alpha^\beta dx \int_\alpha^\beta dx' [h^\dagger(x, x')]^* g_2^*(x') f_1(x)$$

$$[h^\dagger(x, x')]^* = h(x', x)$$
$$h^\dagger(x, x') = h^*(x', x) \quad \textbf{Conjugate transpose}$$

Adjoint of a C-D mapping

For an operator $\mathcal{H}$ mapping from a continuous space to a discrete space $\mathbb{V}$:

$$\mathcal{H} : \mathbb{L}_2(\alpha, \beta) \to \mathbb{E}^M$$

The adjoint of $\mathcal{H}$ is given by:

$$\mathcal{H}^\dagger : \mathbb{E}^M \to \mathbb{L}_2(\alpha, \beta)$$

The adjoint satisfies the inner product relationship:

$$\langle \mathbf{g}_2, \mathcal{H}\mathbf{f}_1 \rangle_{\mathbb{V}} = \langle \mathcal{H}^\dagger \mathbf{g}_2, \mathbf{f}_1 \rangle$$

🧠 *Recall the inner product in a continuous form* (in Lecture 2):
$\langle \mathbf{f}_1, \mathbf{f}_2 \rangle = \int_\alpha^\beta dx f_1^*(x) f_2(x)$

Derivation of the Adjoint

$$\langle \mathbf{g}_2, \mathcal{H}\mathbf{f}_1 \rangle_{\mathbb{V}} = \langle \mathcal{H}^\dagger \mathbf{g}_2, \mathbf{f}_1 \rangle$$

📌 *This derivation is left as an execercise in Homework 2.*

Projection Operators

- Idempotency: Applying the same operator (here, $\mathcal{P}$) multiple times has no further effect:

$$\mathcal{P}^2 = \mathcal{P}$$

- Projection operator (projector): An idempotent operator that is also Hermitian (a square matrix, real numbers, and $\mathcal{P}^T = \mathcal{P}$):

$$\mathcal{P}^\dagger = \mathcal{P}$$

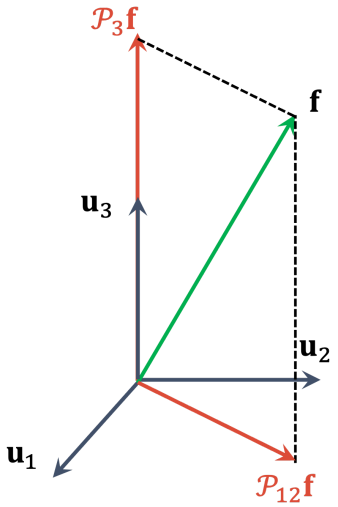
Example: Projection onto a 1D Subspace

Consider a vector decomposition: $\mathbf{f} = \sum_{n=1}^N \alpha_n \mathbf{u}_n$, where $\{\mathbf{u}_n, n = 1, \ldots, N\} \in \mathbb{E}^N$ is the orthonormal basis.

Projection of $\mathbf{f}$ along $\mathbf{u}_n$ or onto the 1D subspace spaned by $\mathbf{u}_n$:

$$\mathcal{P}_n \mathbf{f} = \alpha_n \mathbf{u}_n$$
$$\mathcal{P}_n \mathcal{P}_n \mathbf{f} = \alpha_n \mathbf{u}_n$$

➡️ $\mathcal{P}_n$ sets all components of $\mathbf{f}$ to zero except for the one in the direction of $\mathbf{u}_n$.

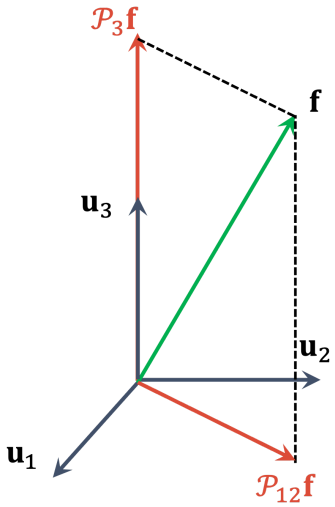


Example: Projection onto a 2D Subspace

Projection of $\mathbf{f}$ onto the nm-plane, i.e., the 2D subspace spanned by $\mathbf{u}_n$ and $\mathbf{u}_m$:

$$\mathcal{P}_{nm}\mathbf{f} = \alpha_n \mathbf{u}_n + \alpha_m \mathbf{u}_m$$
$$\mathcal{P}_{nm}\mathcal{P}_{nm}\mathbf{f} = \alpha_n \mathbf{u}_n + \alpha_m \mathbf{u}_m$$

➡️ $\mathcal{P}_{nm}$ sets all components of $\mathbf{f}$ to zero except those in the directions of $\mathbf{u}_n$ and $\mathbf{u}_m$.



Outer Product Representation

One specific form of projection operator is $\mathcal{P} = \mathbf{p}\mathbf{p}^\dagger$, where $\mathbf{p}$ is a unit-norm vector ($\|\mathbf{p}\|_2 = 1$).

$$\mathcal{P}\mathbf{f} = \mathbf{p}\mathbf{p}^\dagger\mathbf{f} = \mathbf{p}\langle \mathbf{p}, \mathbf{f}\rangle$$

$$\mathcal{P}^2\mathbf{f} = \mathbf{p}\mathbf{p}^\dagger\mathbf{p}\mathbf{p}^\dagger\mathbf{f} = \mathbf{p}\langle \mathbf{p}, \mathbf{p}\rangle\langle \mathbf{p}, \mathbf{f}\rangle = \mathbf{p}\langle \mathbf{p}, \mathbf{f}\rangle = \mathcal{P}\mathbf{f} \quad \textbf{Idempotency}$$

🧠 *Recall the inner product of orthonormal vectors* (in Lecture 3): $\langle \mathbf{p}_n, \mathbf{p}_m\rangle = \delta_{nm}$, i.e., $\langle \mathbf{p}_n, \mathbf{p}_m\rangle = 1$, if $n = m$ and $\langle \mathbf{p}_n, \mathbf{p}_m\rangle = 0$, if $n \neq m$

A general form of a projection operator: $\mathcal{P} = \sum_n \mathbf{p}_n\mathbf{p}_n^\dagger$, where $\{\mathbf{p}_n\}$ is any set of mutually orthogonal vectors with unit norm ($\|\mathbf{p}_n\|_2 = 1$)

$$\mathcal{P}\mathbf{f} = \sum_n \mathbf{p}_n\mathbf{p}_n^\dagger\mathbf{f} = \sum_n \mathbf{p}_n\langle \mathbf{p}_n, \mathbf{f}\rangle$$

$$\mathcal{P}^2\mathbf{f} = \sum_n \mathbf{p}_n\mathbf{p}_n^\dagger \sum_m \mathbf{p}_m\mathbf{p}_m^\dagger\mathbf{f} = \mathbf{p}_n\langle \mathbf{p}_n, \mathbf{p}_m\rangle\langle \mathbf{p}_m, \mathbf{f}\rangle$$
$$= \sum_n \mathbf{p}_n\langle \mathbf{p}_n, \mathbf{f}\rangle = \mathcal{P}\mathbf{f} \quad \textbf{Idempotency}$$

Example: Orthonormal Basis and Identity Operator

Let $\mathbf{f} = \sum_{n=1}^3 \alpha_n \mathbf{u}_n$, where $\mathbf{u}_1 = (1, 0, 0)$, $\mathbf{u}_2 = (0, 1, 0)$, and $\mathbf{u}_3 = (0, 0, 1)$

$$\mathbf{u}_1\mathbf{u}_1^\dagger = \begin{bmatrix}1\\0\\0\end{bmatrix}\begin{bmatrix}1 & 0 & 0\end{bmatrix} = \begin{bmatrix}1 & 0 & 0\\0 & 0 & 0\\0 & 0 & 0\end{bmatrix} \quad \mathbf{u}_2\mathbf{u}_2^\dagger = \begin{bmatrix}0\\1\\0\end{bmatrix}\begin{bmatrix}0 & 1 & 0\end{bmatrix} = \begin{bmatrix}0 & 0 & 0\\0 & 1 & 0\\0 & 0 & 0\end{bmatrix}$$

$$\mathbf{u}_3\mathbf{u}_3^\dagger = \begin{bmatrix}0\\0\\1\end{bmatrix}\begin{bmatrix}0 & 0 & 1\end{bmatrix} = \begin{bmatrix}0 & 0 & 0\\0 & 0 & 0\\0 & 0 & 1\end{bmatrix} \quad \sum_{n=1}^{3}\mathbf{u}_n\mathbf{u}_n^\dagger = \begin{bmatrix}1 & 0 & 0\\0 & 1 & 0\\0 & 0 & 1\end{bmatrix} = \mathbf{I}$$

This demonstrate the **completeness** of the set $\{\mathbf{u}_n\}$, which is also referred to as the **closure** of the set or the **decomposition of the unit operator (identity operator)**.

If the set $\{\mathbf{u}_n\}$ is complete, it satisfies:

$$\sum_{n=1}^{N}\mathbf{u}_n\mathbf{u}_n^\dagger = \mathcal{I}_{\mathbb{U}},$$

where $\mathcal{I}_{\mathbb{U}}$ is the identity operator in $\mathbb{U}$.

Any vector $\mathbf{f} \in \mathbb{U}$ can thus be expressed as:

$$\mathbf{f} = \sum_{n=1}^{N}\mathbf{u}_n\mathbf{u}_n^\dagger\mathbf{f} = \sum_{n=1}^{N}\mathbf{u}_n\langle\mathbf{u}_n,\mathbf{f}\rangle = \sum_{n=1}^{N}\langle\mathbf{u}_n,\mathbf{f}\rangle\mathbf{u}_n = \sum_{n=1}^{N}\alpha_n\mathbf{u}_n$$

🧠 *Recall the expansion coefficient α_m of $\mathbf{f}$ for the basis vector $\mathbf{u}_m$* (in Lecture 3): $\alpha_m = \langle\mathbf{u}_m,\mathbf{f}\rangle$.

Orthogonal Complement and Subsapce Decomposition

Consider the following projectors:

$$\mathcal{P}_3 = \begin{bmatrix}0 & 0 & 0\\0 & 0 & 0\\0 & 0 & 1\end{bmatrix} = \mathcal{P}_{\mathbb{S}}, \quad \mathcal{P}_{12} = \begin{bmatrix}1 & 0 & 0\\0 & 1 & 0\\0 & 0 & 0\end{bmatrix} = \mathcal{P}_{\mathbb{S}_\perp}$$

- $\mathcal{P}_{\mathbb{S}}$: Projector onto the subspace $\mathbb{S}$
- $\mathcal{P}_{\mathbb{S}_\perp}$: Projector onto the subspace $\mathbb{S}_\perp$
- Two subspaces $\mathbb{S}$ and $\mathbb{S}_\perp$ comprise the entire space.
- Two subspaces $\mathbb{S}$ and $\mathbb{S}_\perp$ are orthogonal.
- The subspace $\mathbb{S}_\perp$ is the orthogonal complement of $\mathbb{S}$.

Consider the following projectors:

$$\mathcal{P}_3 = \begin{bmatrix}0 & 0 & 0\\0 & 0 & 0\\0 & 0 & 1\end{bmatrix} = \mathcal{P}_{\mathbb{S}}, \quad \mathcal{P}_{12} = \begin{bmatrix}1 & 0 & 0\\0 & 1 & 0\\0 & 0 & 0\end{bmatrix} = \mathcal{P}_{\mathbb{S}_\perp}$$

The two projection operators are orthogonal complements:

$$\mathcal{P}_{\mathbb{S}_\perp} = \mathcal{I} - \mathcal{P}_{\mathbb{S}}$$

Thus, any vector $\mathbf{f}$ satisfies:

$$\mathbf{f} = \mathcal{P}_{\mathbb{S}}\mathbf{f} + \mathcal{P}_{\mathbb{S}_\perp}\mathbf{f}$$
$$\mathcal{I}\mathbf{f} = \mathcal{P}_{\mathbb{S}}\mathbf{f} + \mathcal{P}_{\mathbb{S}_\perp}\mathbf{f}.$$

